CS 6012: Autumn 2026 — Open Project

Updated Schedule & Milestone Tracking Spreadsheet (as of October 25, 2026)

Team: Aayush Gupta (25m0140) & Srijal Gupta (25m0387) | **Project:** GNN Molecular Substructure Search **Live Online:** docs.google.com/spreadsheets/d/1264hAmtv7T5bGPyuDbSqR52UGxm6RZCEa7NFxWk-cQ4

#	Milestone / Task Description	Target Date / Window	Status	Deliverable / Technical Remarks
1	**Literature Survey & Reference Paper Study**	Oct 01 – Oct 08, 2026	COMPLETED	Surveyed NeuroMatch (ICLR 2020), GIN (ICLR 2019), and PharmacoMatch (ICLR 2025).
2	**Problem Brainstorming & Architecture Scoping**	Oct 05 – Oct 09, 2026	COMPLETED	Decided on two-stage cascade: GNN fast screening + exact RDKit verification.
3	**Project Proposal Form Submission**	Sat, Oct 10, 2026 (11:59 PM)	COMPLETED	Form submitted on Google Forms with problem statement, datasets, and team roles.
4	**Baseline Setup & RDKit Fingerprint Implementation**	Oct 11 – Oct 14, 2026	COMPLETED	Implemented 2048-bit Chem.PatternFingerprint screen (zero false negatives baseline).
5	**Teaching Team Review & Disqualification Window**	Wed, Oct 14, 2026	COMPLETED	Proposal officially accepted with no revision or disqualification flagged.
6	**Updated Proposal Submission Window**	Sat, Oct 17, 2026	COMPLETED	Not required (original proposal cleared all course checks).
7	**Software Design & Graph Featurization Module**	Oct 12 – Oct 16, 2026	COMPLETED	Atom featurization (10-dim one-hot) and bond featurization (4-dim one-hot) for PyG.
8	**Data Collection & Partitioning (ZINC250k)**	Oct 14 – Oct 17, 2026	COMPLETED	Loaded ZINC250k: 40k train, 4k val, 50k DB, 2k queries. Disjoint splits established.
9	**Query Generation & Hard Negative Mining**	Oct 15 – Oct 18, 2026	COMPLETED	Extracted 4–12 atom subgraphs via BFS; mined hard negatives failing exact match.
10	**Intermediate Checkpoint #1: Submission 1 (Position Paper)**	Sun, Oct 18, 2026 (11:59 PM)	COMPLETED	Submitted 1-page PDF status report on Responsible Healthcare DL to Moodle.
11	**Intermediate Checkpoint #2: Submission 2 (Code Project)**	Mon, Oct 19, 2026 (11:59 PM)	COMPLETED	Submitted 1-page PDF status report with initial pipeline and GitHub progress.
12	**Model Training & Order-Embedding Loss Optimization**	Oct 20 – Oct 23, 2026	COMPLETED	Trained Model A (GIN+MLP, BCE) & Model B (NeuroMatch order margin loss) on GPU.
13	**Freeze Core Model Code & Pilot Result Generation**	Oct 22 – Oct 24, 2026	COMPLETED	Pilot 5,000 DB benchmark evaluated. Saved benchmark CSVs, ROC curves & tradeoff plots.
14	**Position Paper Final Submission (Slides PDF + Video)**	Sun, Oct 25, 2026 (11:59 PM)	COMPLETED	10-minute presentation recorded by both members (<30MB) and uploaded to Moodle.
15	**Intermediate Checkpoint #4: Code Project Status Report**	Sun, Oct 25, 2026 (11:59 PM)	IN PROGRESS	Submitting 1-page status report PDF + 2nd companion PDF of updated schedule.
16	**Codebase Refactoring & Colab Documentation**	Oct 26 – Oct 29, 2026	IN PROGRESS	Polishing inline docstrings, README instructions, and interactive Colab demo.
17	**Full 50k Database Benchmark & Error Analysis**	Oct 28 – Nov 01, 2026	SCHEDULED	Scaling cascade evaluation to 50k molecules across query atom strata (4–12 atoms).
18	**Final Project Report Document Drafting**	Oct 29 – Nov 03, 2026	SCHEDULED	Writing comprehensive report: methodology, order embeddings, Pareto tradeoff & findings.
19	**Results Video Recording (<= 2 minutes limit)**	Nov 01 – Nov 03, 2026	SCHEDULED	Recording concise demonstration video showing code execution and benchmark plots.
20	**Final Submission: Submission 2 Due (Code, Report, Video)**	Wed, Nov 04, 2026 (11:59 PM)	SCHEDULED	Final submission on Moodle and GitHub repository (Viva defense Nov 7–8).